



MAY 6, 2026

OC: SE FORMATIVE ASSESSMENT 2

NON-FUNCTIONAL TESTING DOCUMENTATION

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1. Performance Testing

Report Loading Time

Method:

The report generation feature was tested by timing the duration from clicking the 'Generate Report' button to the report being fully displayed on screen. Testing was conducted using the Visual Studio Diagnostic Tools and a manual stopwatch across 4 iterations.

Results:

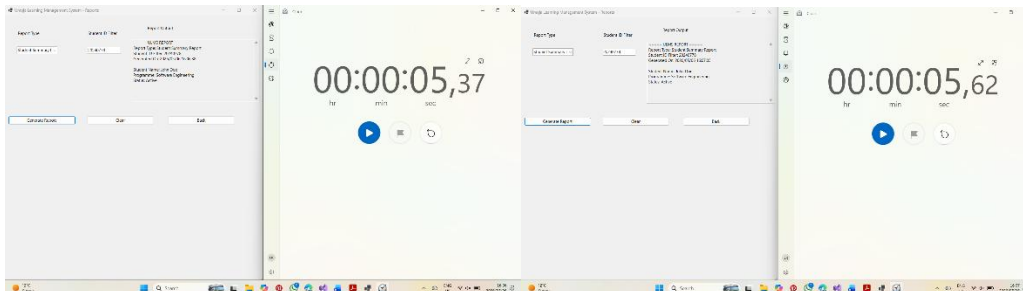
Run	Action	Threshold	Time Taken	Result
Run 1	Load Student Summary Report	< 3 sec	5.37 sec	FAIL
Run 2	Load Student Summary Report	< 3 sec	5.62 sec	FAIL
Run 3	Load Student Summary Report	< 3 sec	5.62 sec	FAIL
Run 4	Load Student Summary Report	< 3 sec	5.41 sec	FAIL

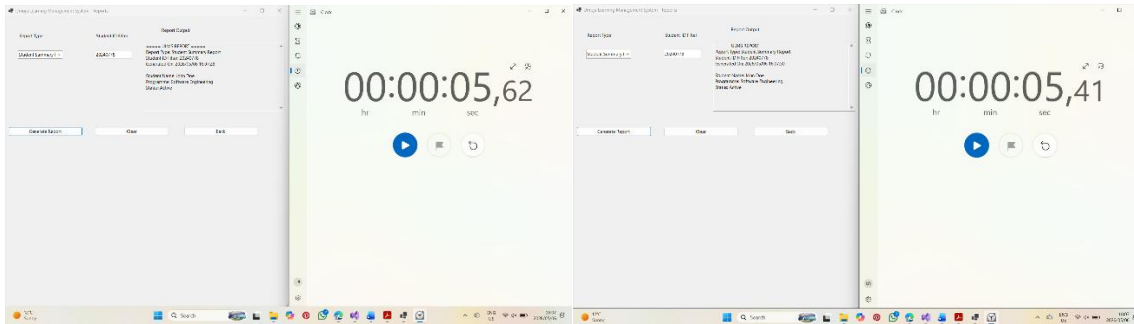
Interpretation:

The system failed to meet the 3 second threshold in all 4 iterations. The system has an average loading time of 5.51 seconds based on the results. Due to the consistency of the results of the four iterations, this confirms that the issue embedded in the report logic itself rather than external issues such as network issues.

Recommendation: The development team needs to investigate the database query on the report generation module and implement SQL indexing on the relevant student and marks tables.

Evidence:





2. Usability Testing

Navigation Steps

Method:

User journeys are evaluated by counting the number of mouse clicks or keyboard actions required to complete each task. An industry benchmark of 5 or fewer steps for any core task was used as the acceptance criterion.

Results:

Task	Steps Required	Benchmark	Result
Login and reach Dashboard	4	≤5	PASS
Register a new student	5	≤5	PASS
Capture a mark for a student	4	≤5	PASS
Generate a Summary Report	4	≤5	PASS

Interpretation:

All four core user journeys were completed within the accepted industry benchmark of 5 steps or fewer, resulting in a full pass across all usability test tasks.

Please note the counting methodology applied during this test. Filling in all details within a form, including dropdown menu selections, was counted as a single consolidated step, reflecting the user's experience of completing one logical action rather than counting each individual field. And also, clicking the back button to return to the previous screen was counted as a discrete step, since the application does not automatically redirect the user back to the Dashboard upon task completion.

Recommendation: While the system passes the usability benchmark under the current counting method, the development team should consider implementing automatic redirection to the Dashboard or a logical next screen upon successful task completion. This would further improve the user experience, bringing the ULMS application closer to professional-grade standards.

3. Security Testing

Invalid Login Attempts

Method:

Here, the system will be tested to evaluate whether it can protect itself against repeated failed login attempts and whether it handles edge case inputs securely.

Results:

Test	Input	Expected Behaviour	Actual Output	Result
3 wrong passwords	Wrong password × 3	Error each time	“Login successful” message shown	FAIL
5 wrong attempts (lockout)	Wrong password × 5	Account locked	No lockout occurs and “Login successful” message shown	FAIL
SQL Injection attempt	' OR '1'='1	Rejected	System displayed "Invalid login credentials." and denied access	PASS
Empty fields	Blank username or password	Validation message	"Invalid login credentials" shown	FAIL

Interpretation:

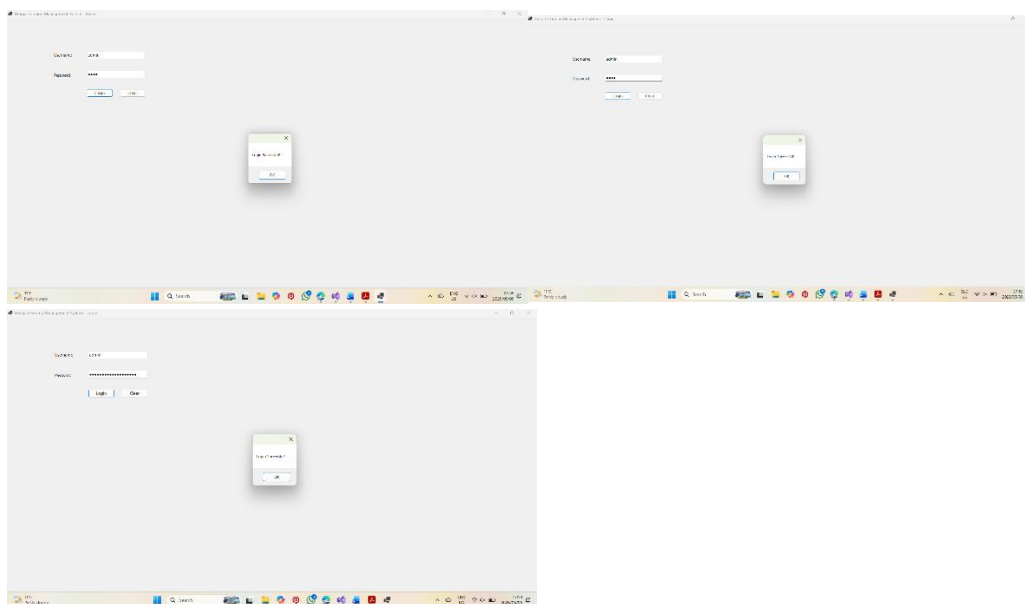
The system failed 3 out of 4 security test cases. The SQL injection attempt was successfully blocked but the system allowed access after multiple incorrect passwords and failed to lock the account after five consecutive failed attempts. Additionally, entering empty fields did not trigger the expected specific validation messages.

The SQL injection PASS shows that the application is likely utilizing parameterized queries or simple string comparison for authentication rather than raw SQL concatenation.

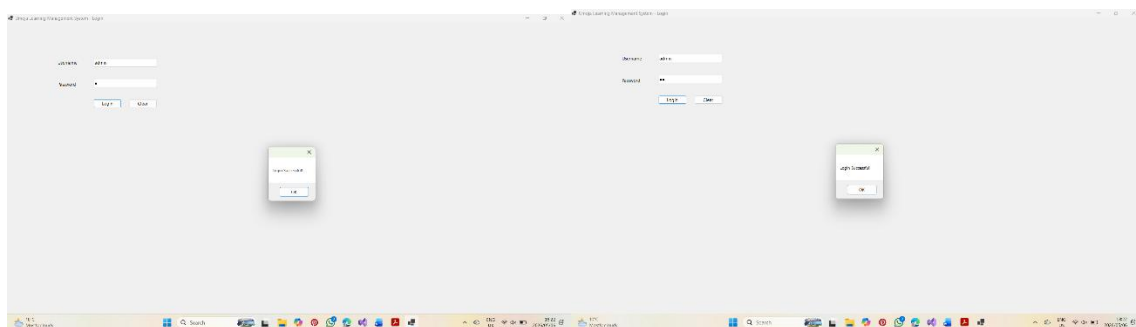
Recommendation: Before deployment, the developers should repair the authentication logic, implement account lockout and enhance the current input validation.

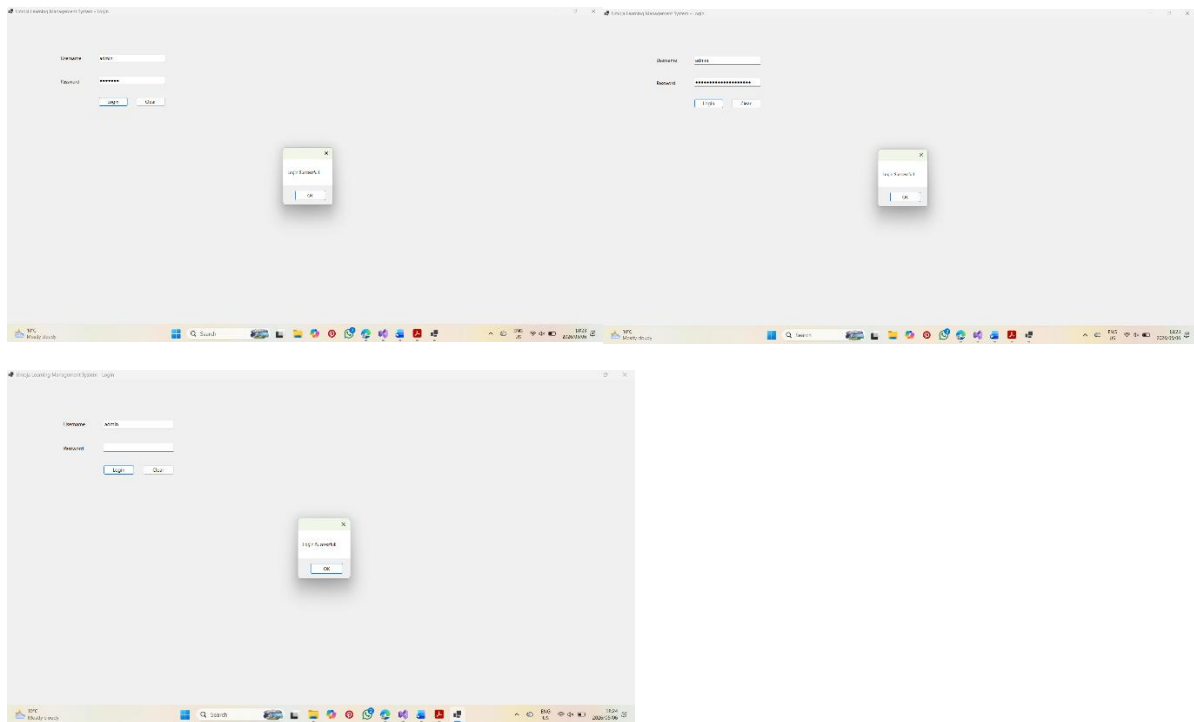
Evidence:

1. Wrong Password × 3

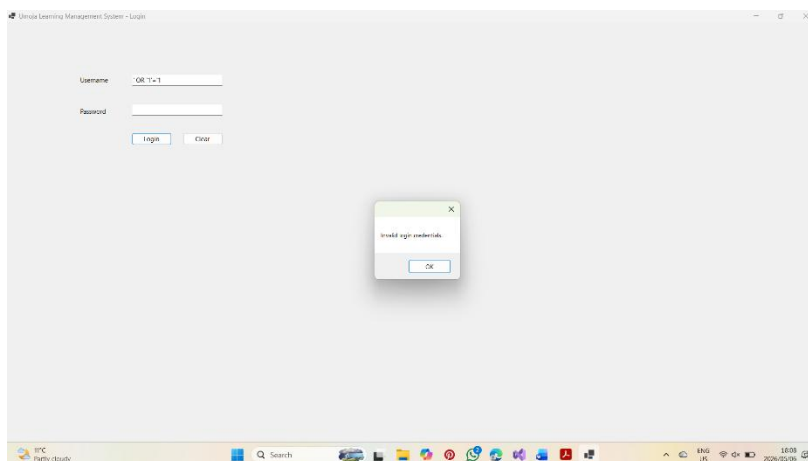


2. Account lockout – wrong password × 5

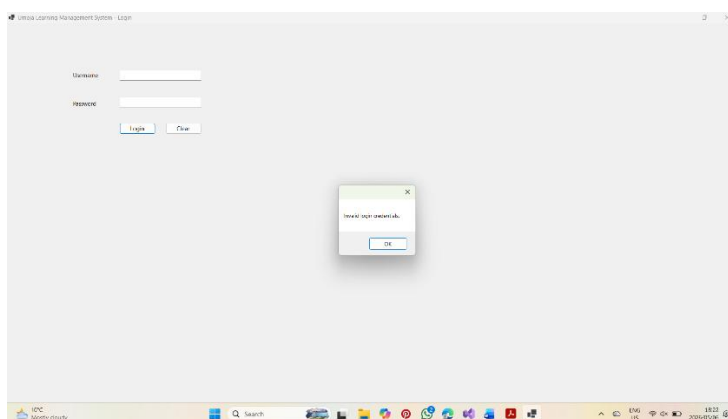




3. SQL Injection Attempt



4. Empty fields



4. Reliability Testing

Repeated Operations

Method:

Here, the same operation is running multiple times consecutively to check whether the system behaves consistently, or it crashes or produces different results over time.

Results:

Operation	Repetitions	Consistent Result	Observation
Valid Login	5	Yes	PASS — System responded consistently each time
Student Registration	5	Yes	PASS — Records saved without errors
Marks capture and average	5	No	FAIL — Average result inconsistent across repetitions; defect confirmed

Interpretation:

Login and registration are reliable. The marks capture module is unreliable due to the confirmed average calculation defect identified in TC#006 through TC#009. The system cannot be considered reliable until BUG#004 is resolved and average calculations produce consistent results across all repetitions.